

For some, the 5.1 percent real return might be a properly adjusted starting point for a portfolio return assumption to project retirement outcomes. We are getting closer. However, there are still a number of further adjustments we must consider to provide realistic expectations about what may happen in retirement, such as incorporating taxes on interest and dividends, accounting for the fact that today's interest rates are much lower than historical averages, accounting for the impact of investment management fees, and considering the possibility of outperformance or underperformance with respect to the underlying market indexes.

Also of utmost importance is making a downward adjustment to increase the probability that the assumed return will be met. For a best guess about the forward-looking compounded real return, there is still a 50 percent chance that the actual realized return will be lower. Retirees generally focus on building a plan with a high probability of success, which implicitly means they must assume a lower return than average. How should we adjust our return assumptions to account for this aspect of risk? This is an important matter that we will revisit in the next chapter.

For the time being, Exhibit 2.1 does also include a further adjustment for today's low interest rate environment. With a yield to maturity of 0.1 percent on five-year Treasury Inflation-Protected Securities (TIPS) at the start of 2017, the real interest rate as I write is about 2.1 percent less than the historical average compounded real return on intermediate-term government bonds. Accounting for lower interest rates but maintaining the same historical premium that stocks have earned above bonds, the exhibit provides an arithmetic real return of 3.5 percent and a compounded real return of 2.7 percent for a 50/50 portfolio with the same historical volatility. These returns are much less than 8 to 12 percent, and they should serve as the anchor for choosing a realistic fixed-return assumption.